RES-GEO — Barrier Geometry and Spatial Representation

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Table of Contents

1 Domain Purpose and Scope

RES-GEO defines the geometry taxonomy, descriptors and software representation needed for fixed-rigid barrier-class comparison in the Kamaishi and Sendai local tsunami corridors. The active classes are wall-type barriers and obstacle, dissipating or array-type barriers.

The current Research scope is not unrestricted geometry generation. Existing or reconstructed defences may be included where source data are sufficient, but the baseline comparison is between evidence-backed barrier classes with explicit dimensions, placement rules and numerical representation. ML-driven, surrogate-driven and generative geometry work remains deferred until the hydrodynamic data pipeline and validation hierarchy are accepted.

2 Domain Deliverables

3 RES-GEO-TAX — Barrier Geometry Taxonomy and Candidate Families

3.1 Purpose and Scope

This Deliverable classifies the active fixed-rigid barrier families used for the Tohoku corridor comparison. It admits two baseline families:

1. wall-type barriers, including continuous or segmented seawalls, ridges, breakwaters and raised impermeable lines;
2. obstacle, dissipating or array-type barriers, including finite blocks, columns, staggered rows, permeable-looking resolved openings and other structures whose effect is controlled by blockage, spacing and wake interaction.

Selection is based on geometric recoverability, hydrodynamic relevance and numerical representability. The deliverable excludes unrestricted geometry generation and does not treat damage, scour or deformable barrier response as active baseline geometry.

3.2 Research Questions

1. Which barrier families are active in the Kamaishi and Sendai comparison cases?

2. Which minimum descriptors distinguish wall-type and obstacle/array-type barriers for the local URANS–VOF model?

3. Which real-defence geometries have enough provenance to be reconstructed rather than treated as TODO candidates?

3.3 Terminology and Definitions

Wall-type barrier A laterally extended defence whose principal hydrodynamic effect is reflection, run-up, overtopping and blocking along a near-continuous crest.

Obstacle or dissipating barrier A finite-width body or row of bodies whose principal hydrodynamic effect is local blockage, separation, wake production, transmission and turbulent energy loss.

Array A repeated set of resolved obstacle elements with spacing, staggering and row count recorded explicitly.

Fixed rigid barrier A solid geometry that supplies no-slip and no-penetration wall patches to the local fluid solver and does not deform during the baseline hydrodynamic calculation.

3.4 Literature Evidence

del Jesus, Lara, and Losada [1] supports the taxonomy split between impermeable vertical barriers and barriers whose porosity or openings require explicit modelling assumptions. Source role: supports the wall-type versus dissipating/opening class distinction and limits any claim about unresolved porous resistance. Project conclusion: porosity-like behaviour is active only when openings are resolved by geometry or a later porous closure is formally activated.

Watanabe et al. [2] supports using controlled idealised coastal structures for local hydrodynamic validation. Source role: supports simple fixed obstacle and wall geometries as validation-facing families rather than arbitrary generated forms. Limitation: the paper does not define the final Kamaishi or Sendai defence geometry.

Mori, Takahashi, and The 2011 Tohoku Earthquake Tsunami Joint Survey Group [3] supports retaining Sanriku/Kamaishi and Sendai coastal-plain settings as contrasting Tohoku impact regimes. Source role: motivates site-facing geometry classes, not detailed barrier dimensions. TODO/missing-source: identify a citable primary source for the Kamaishi bay-mouth breakwater geometry, 2011 damage state and any reconstruction dimensions before recreating it as a project barrier.

3.5 Governing Relations and Quantitative Definitions

For software-facing comparison, each geometry record shall include a class label and a minimum descriptor vector:

$$\mathcal{G}_j = \{c_j,\, \mathbf{x}_j,\, \psi_j,\, H_j,\, B_j,\, L_j,\, C_j,\, S_j,\, N_j,\, \chi_j\} \tag{3.1}$$

where c_j is the barrier class, $\mathbf{x}_j$ is the placement reference point in the local corridor coordinate system, ψ_j is the plan-view orientation, H_j is height, B_j is width or thickness, L_j is along-barrier length, C_j is crest elevation, S_j is spacing for arrays, N_j is element or row count and χ_j records resolved openings, roughness and provenance flags. Decision role: Equation (3.1) is a data contract, not a design-generation equation. Supporting evidence: del Jesus, Lara, and Losada [1] and Watanabe et al. [2]. Limitation: final admissible parameter bounds remain site- and mesh-dependent.

3.6 Geometry Family or Representation

The active family set is:

- `wall`: continuous or segmented impermeable line defence, with optional resolved openings;
- `obstacle`: isolated block, pier, mound or short barrier element;
- `array`: repeated obstacle elements whose spacing, staggering and row count are explicit;
- `dissipating`: resolved obstacle or array geometry intended to increase separation, wake loss, reflection or overtopping reduction.

Unresolved porous-media barriers, compliant structures, scour-adaptive foundations and damage-evolving barriers are deferred extensions.

3.7 Design Variables and Parameter Bounds

3.8 Geometric Descriptors

3.9 Placement and Arrangement Rules

3.10 Feasibility Constraints

3.11 Two- and Three-Dimensional Representation

3.12 Candidate Approaches

3.13 Critical Comparison

3.14 Assumptions and Applicability

3.15 Limitations and Failure Conditions

3.16 Project-Specific Conclusions

The project taxonomy is now class-comparison led. A geometry may enter the active campaign when it has a traceable descriptor record, a fixed rigid local representation and a clear hydrodynamic purpose. Real defence recreation is allowed only after source geometry is identified; otherwise the case remains a TODO rather than an invented citation.

3.17 Selected Approach and Rationale

The selected approach is a small active taxonomy: wall-type barriers and obstacle/dissipating/array barriers. The rejected active approach is unrestricted generated geometry, because the present work must first stabilise data handling, local hydrodynamic load extraction and the V1–V2–V3 validation hierarchy.

3.18 Unresolved Decisions

- TODO: obtain a primary or peer-reviewed source for Kamaishi bay-mouth breakwater dimensions and 2011 condition.
- Select final descriptor bounds after RES-DAT-GEO terrain extraction and RES-NUM-MSH mesh constraints are available.
- Decide whether any dissipating class requires a porous closure rather than resolved openings.

3.19 Requirements and Handoffs

Software shall support a geometry manifest containing the descriptor record in Equation (3.1), class labels, local corridor coordinates, source/provenance fields and flags for resolved openings versus deferred porous physics. ‘RES-MOD-IBC‘ and ‘RES-NUM-MSH‘ shall consume this record as fixed terrain/barrier wall patches; ‘RES-VER-MET‘ shall use the same identifiers for load and energy metrics.

3.20 Deliverable Completion Record

4 RES-GEO-PAR — Parametric Geometry and Design-Variable Definition

4.1 Purpose and Scope

This Deliverable defines the bounded parameters required to compare fixed-rigid barrier classes in the two local corridors. It does not define an optimisation-ready free-form design space and does not activate ML or generative geometry.

4.2 Research Questions

1. Which geometric parameters are required for wall-type and obstacle/array-type barriers?
2. Which parameters are measured, inferred, scenario-controlled or unresolved?
3. Which parameter bounds are controlled by terrain coverage, corridor width and meshability rather than design preference?

4.3 Terminology and Definitions

4.4 Literature Evidence

del Jesus, Lara, and Losada [1] supports separate parameter treatment for impermeable walls and barriers whose openings or porosity change hydraulic response. Source role: supports explicit opening/spacing parameters and limits unresolved porous claims. Watanabe et al. [2] supports simple controlled geometry parameters for local impact and loading validation. Project conclusion: the parameter set shall remain small, inspectable and tied to hydrodynamic metrics.

4.5 Governing Relations and Quantitative Definitions

For each case j, the comparison parameter vector is:

$$\boldsymbol{\theta}_j^{\text{geo}} = \left[H_j,\, C_j,\, B_j,\, L_j,\, \psi_j,\, x'_j,\, y'_j,\, S_j,\, N_j,\, \phi_j^{\text{open}}\right]^{\mathsf{T}} \tag{4.1}$$

where ϕ_j^{open} is the resolved open-area or gap fraction for array or segmented cases. Source role: del Jesus, Lara, and Losada [1] supports separating impermeable and opening-controlled barriers; Watanabe et al. [2] supports maintaining geometry parameters that map directly to local loading observables. Limitation: Equation (4.1) does not set final bounds.

4.6 Geometry Family or Representation

4.7 Design Variables and Parameter Bounds

Parameter sets shall distinguish measured geometry, inferred geometry, scenario-controlled geometry and deferred structural assumptions. Candidate parameters may include crest height, barrier height, wall or element thickness, footprint length, plan orientation, corridor placement, gap spacing, row count, resolved opening fraction and surface roughness flag. Foundation strengthening, material changes, damage evolution and deformable response are stretch extensions owned by RES-STR after hydrodynamic load extraction is stable.

4.8 Geometric Descriptors

4.9 Placement and Arrangement Rules

4.10 Feasibility Constraints

4.11 Two- and Three-Dimensional Representation

4.12 Candidate Approaches

4.13 Critical Comparison

4.14 Assumptions and Applicability

4.15 Limitations and Failure Conditions

4.16 Project-Specific Conclusions

The active Studentship comparison requires a compact parameter vector, not an open-ended design grammar. Geometry choices must be reproducible from the manifest, meshable in the local domain and traceable to barrier-performance metrics.

4.17 Selected Approach and Rationale

The selected approach is a descriptor-first parameter set for fixed wall, obstacle, dissipating and array barriers. Broad geometric optimisation and generative parameter discovery are formally deferred.

4.18 Unresolved Decisions

- Final numeric bounds for H_j, B_j, L_j, S_j and N_j remain open until corridor terrain and mesh constraints are accepted.
- Real-defence reconstruction parameters for Kamaishi remain TODO pending a citable geometry source.
- Roughness and opening/porosity flags remain provisional until RES-MOD-SRC decides whether any porous closure is active.

4.19 Requirements and Handoffs

Software shall ingest Equation (4.1) from the geometry manifest, generate fixed barrier patches, and preserve stable case identifiers for metric extraction. RES-NUM-MSH shall report whether each parameter set is meshable; RES-VER-MET shall record which geometry parameters produced each run-up, overtopping, force, moment and energy metric.

4.20 Deliverable Completion Record

5 RES-GEO-MET — Geometric and Hydraulic Descriptor Framework

5.1 Purpose and Scope

This Deliverable defines the minimum geometric and hydraulic descriptors needed to compare wall-type and obstacle/array-type barriers in fixed-rigid local tsunami simulations. Descriptor records shall preserve provenance and uncertainty. Damage state, material deformation and scour descriptors are recorded only as deferred structural extensions.

5.2 Research Questions

1. Which descriptors are required before local hydrodynamic comparison can begin?
2. Which descriptors drive software geometry, and which are post-processing metrics emitted by the solver?
3. Which descriptors must carry uncertainty/provenance flags?

5.3 Terminology and Definitions

5.4 Literature Evidence

del Jesus, Lara, and Losada [1] links vertical barrier form, porosity and transmission to hydraulic response. Source role: supports including blockage, opening and transmission descriptors. Watanabe et al. [2] links local geometry to separate force, pressure, velocity and free-surface validation quantities. Project conclusion: descriptors must identify both what was simulated and which hydrodynamic outputs are meaningful for comparison.

5.5 Governing Relations and Quantitative Definitions

For array and obstacle cases, the plan blockage ratio shall be reported as:

$$\beta_{\text{blk}} = \frac{A_{\perp,\text{solid}}}{A_{\perp,\text{corridor}} + \epsilon_A} \tag{5.1}$$

where $A_{\perp,\text{solid}}$ is the projected solid area normal to the corridor centreline and $A_{\perp,\text{corridor}}$ is the available projected corridor area at the comparison section. Decision role: Equation (5.1) is a comparison descriptor for wall/obstacle/array cases. Supporting evidence: del Jesus, Lara, and Losada [1]. Limitation: it does not replace full 3D flow solution because wake, overtopping and pressure depend on local geometry and bathymetry.

5.6 Geometry Family or Representation

5.7 Design Variables and Parameter Bounds

5.8 Geometric Descriptors

Required pre-simulation descriptors are class label, local corridor coordinates, orientation, crest elevation, height, thickness or element width, footprint length, spacing, row count, opening fraction, roughness flag, blockage ratio and provenance quality.

Required post-simulation hydraulic descriptors are maximum run-up, overtopping volume or discharge, transmitted and reflected energy measures, downstream maximum depth and velocity, peak pressure, resultant force, impulse and moment. Damage, survival state and structural-capacity proxies are retained only for later RES-STR screening and shall not be used to claim baseline hydrodynamic performance.

5.9 Placement and Arrangement Rules

5.10 Feasibility Constraints

5.11 Two- and Three-Dimensional Representation

5.12 Candidate Approaches

5.13 Critical Comparison

5.14 Assumptions and Applicability

5.15 Limitations and Failure Conditions

5.16 Project-Specific Conclusions

The geometry descriptor framework is sufficient when each barrier case can be reconstructed, meshed, run and post-processed using stable identifiers. It is not sufficient for structural certification or damage prediction.

5.17 Selected Approach and Rationale

The selected descriptor approach separates pre-simulation geometry fields from solver-emitted hydrodynamic metrics. This keeps software data handling parallel with Research without moving solver implementation into RES-GEO.

5.18 Unresolved Decisions

Open decisions are the final blockage-normalisation section, roughness descriptor values, gap/opening representation and accepted uncertainty classes for inferred geometry.

5.19 Requirements and Handoffs

Software shall emit descriptors and metrics using the same `corridor_id`, `geometry_id` and `case_id`. RES-PHY-BAR and RES-PHY-LOD shall define the hydrodynamic meaning of the emitted metrics; RES-VER-MET shall define acceptance tolerances.

5.20 Deliverable Completion Record

6 RES-GEO-ARR — Barrier Arrangement, Placement and Spatial Configuration

6.1 Purpose and Scope

This Deliverable defines how active barrier classes are placed inside the local corridor coordinate system. Arrangement is constrained by the evidence-based Kamaishi and Sendai corridor polygons, not by an arbitrary drawing canvas. The deliverable excludes global coastline redesign and generative placement optimisation.

6.2 Research Questions

1. How is a barrier positioned relative to the corridor centreline x' and width direction y'?
2. How are continuous walls distinguished from finite obstacles and arrays?
3. Which placement choices are physical scenario decisions, and which are numerical buffer or mesh-quality decisions?

6.3 Terminology and Definitions

x' is the local propagation-direction coordinate supplied by RES-DAT-SCN, y' is the transverse corridor coordinate, and z' is vertical elevation in the solver datum. The angle ψ_j is the plan-view orientation of barrier j measured relative to the x' axis; $\psi_j = \pi/2$ denotes a barrier aligned across the corridor.

6.4 Literature Evidence

Mori, Takahashi, and The 2011 Tohoku Earthquake Tsunami Joint Survey Group [3] supports retaining different placement contexts for Sanriku/Kamaishi and Sendai because the same event produced contrasting run-up and inundation behaviour. Source role: supports corridor-specific placement rather than one generic test site. Watanabe et al. [2] supports preserving exact probe and structure locations for validation-facing local simulations. Limitation: neither source supplies final barrier coordinates; those remain RES-DAT-SCN/RES-DAT-CAS data products.

6.5 Governing Relations and Quantitative Definitions

The plan-view local placement vector is:

$$\mathbf{r}'_j = \left[x'_j,\, y'_j\right]^{\mathsf{T}} \tag{6.1}$$

The local basis for each barrier is:

$$\mathbf{e}_{n,j} = [\cos\psi_j,\, \sin\psi_j]^{\mathsf{T}} \tag{6.2}$$

$$\mathbf{e}_{t,j} = [-\sin\psi_j,\, \cos\psi_j]^{\mathsf{T}} \tag{6.3}$$

Decision role: these definitions make barrier placement reproducible from the corridor transform. Supporting evidence: RES-DAT-SCN owns the corridor construction; Watanabe et al. [2] supports validation-facing control of structure and probe locations.

6.6 Geometry Family or Representation

Continuous walls are represented as elongated patches whose tangent direction is $\mathbf{e}_{t,j}$. Obstacle and array cases are represented as one or more finite elements with recorded centre points, spacing and row index.

6.7 Design Variables and Parameter Bounds

Arrangement variables include x'_j, y'_j, ψ_j, number of rows, along-row spacing, cross-row spacing, stagger offset and gap/opening fraction. These variables are comparison controls, not yet an optimisation search space.

6.8 Geometric Descriptors

Placement descriptors shall include clearance to artificial inlet, outlet and side boundaries; clearance to steep terrain changes; and distance from run-up, overtopping and wake measurement sections.

6.9 Placement and Arrangement Rules

Baseline placement rules are:

1. place wall-type cases approximately transverse to x' unless a real coastline, harbour or defence alignment provides a citable alternative;

2. place array-type cases with all elements inside the accepted terrain/corridor polygon and away from sponge zones;
3. keep comparison sections fixed when comparing barrier classes;
4. do not narrow the corridor to fit a barrier unless the narrowing represents a physical bay or coastline constraint.

6.10 Feasibility Constraints

A placement is infeasible if it intersects missing terrain data, requires a nonphysical wall at the side boundary, violates local mesh quality constraints or places the barrier inside a numerical relaxation region.

6.11 Two- and Three-Dimensional Representation

6.12 Candidate Approaches

6.13 Critical Comparison

6.14 Assumptions and Applicability

6.15 Limitations and Failure Conditions

6.16 Project-Specific Conclusions

For the Studentship campaign, arrangement is part of the data/model interface. It links source-to-coast trajectory, terrain extraction, barrier geometry and validation probes; it is not a separate design art exercise.

6.17 Selected Approach and Rationale

The selected approach is corridor-relative placement using (x', y', z') coordinates and fixed comparison sections. Free placement optimisation is deferred.

6.18 Unresolved Decisions

Open decisions are final Kamaishi and Sendai target coordinates, final comparison sections, sponge widths, and any real-defence alignment source for the Kamaishi breakwater.

6.19 Requirements and Handoffs

Software shall generate barrier placement from corridor coordinates, clip it against accepted terrain/corridor polygons, and emit placement diagnostics for mesh quality and validation probe locations.

6.20 Deliverable Completion Record

7 RES-GEO-REP — Two- and Three-Dimensional Model Representation

7.1 Purpose and Scope

This Deliverable defines how each active fixed-rigid barrier geometry is represented for the regional model, local 3D fluid model, hydrodynamic-load handoff and later deferred structural analysis. ML feature generation is not active; any feature representation is a future consumer of the same manifest.

7.2 Research Questions

1. What is the minimum regional footprint required for regional propagation and replay extraction?
2. What local 3D surface representation is required for no-slip/no-penetration wall patches and load extraction?
3. Which representation fields must be preserved for later structural or ML use without activating those workstreams now?

7.3 Terminology and Definitions

7.4 Literature Evidence

Watanabe et al. [2] supports explicit local geometry, probe and loading-output control in tsunami-impact validation. del Jesus, Lara, and Losada [1] supports distinguishing resolved solid geometry from unresolved porous closure. Project conclusion: the local representation shall be a resolved fixed solid surface unless a separate porous model is later accepted.

7.5 Governing Relations and Quantitative Definitions

The fixed barrier patch in the local fluid domain is:

$$\Gamma_{\mathrm{B}} = \partial\Omega_{\mathrm{f}} \cap \partial\Omega_{\mathrm{B}} \tag{7.1}$$

where Ω_{f} is the local fluid-control domain and Ω_{B} is the solid barrier volume. Decision role: Equation (7.1) defines the geometric surface on which no-slip/no-penetration conditions and hydrodynamic load integration are applied. Supporting evidence: Watanabe et al. [2]. Limitation: it does not describe deformable FSI interfaces.

7.6 Geometry Family or Representation

7.7 Design Variables and Parameter Bounds

7.8 Geometric Descriptors

7.9 Placement and Arrangement Rules

7.10 Feasibility Constraints

7.11 Two- and Three-Dimensional Representation

Representation levels shall include:

- a regional-scale footprint or roughness/obstruction marker, used only if the regional model needs a defence representation;
- a local three-dimensional fixed solid surface for the URANS–VOF barrier interaction model;
- a patch and face identifier set for pressure, shear, force, impulse and moment extraction;
- a deferred structural surface mapping for one-way CFD-to-FEA load transfer after hydrodynamic extraction is stable;
- optional future ML descriptors generated from the same manifest, not an independent geometry source.

7.12 Candidate Approaches

7.13 Critical Comparison

7.14 Assumptions and Applicability

7.15 Limitations and Failure Conditions

7.16 Project-Specific Conclusions

The local solver needs an explicit fixed surface representation before any barrier comparison is meaningful. The same surface identifiers must survive into post-processing so forces and moments can be traced back to geometry cases.

7.17 Selected Approach and Rationale

The selected local representation is a fixed rigid solid volume converted to terrain/barrier wall patches and stable patch identifiers. Moving boundaries, topology change and deformable surfaces are deferred to RES-STR-FSI.

7.18 Unresolved Decisions

Open decisions are the exact geometry file format, wall roughness metadata, CAD/CSG versus raster-derived surface generation, and Kamaishi real-defence source geometry.

7.19 Requirements and Handoffs

Software shall support fixed geometry ingestion, patch labelling, face/patch metric extraction and structured export to HDF5 or an equivalent schema. RES-STR-LOD shall consume the same patch identifiers for later hydrodynamic load transfer.

7.20 Deliverable Completion Record

8 RES-GEO-CST — Geometric Feasibility and Design Constraints

8.1 Purpose and Scope

This Deliverable narrows admissible geometry cases to fixed-rigid barrier-class comparisons that can be represented on the accepted terrain, meshed robustly and interpreted through hydrodynamic outputs. Material limits, foundation constraints, deformability and damage are recorded as stretch structural concerns rather than active geometric feasibility criteria.

8.2 Research Questions

8.3 Terminology and Definitions

8.4 Literature Evidence

8.5 Governing Relations and Quantitative Definitions

8.6 Geometry Family or Representation

8.7 Design Variables and Parameter Bounds

8.8 Geometric Descriptors

8.9 Placement and Arrangement Rules

8.10 Feasibility Constraints

Candidate interventions shall be rejected or marked unresolved if they cannot be constructed plausibly, cannot be represented in the selected numerical workflow, violate known material or foundation constraints, or lack enough provenance to support simulation.

For the active hydrodynamic baseline, a geometry shall also be rejected or deferred if it requires unresolved porous resistance, moving structural boundaries, scour-induced terrain change or material damage to explain its intended performance.

Selected approach. Feasibility is first a data and numerical representation gate: the case must fit within the accepted corridor polygon, avoid sponge zones, preserve mesh quality near walls and have a traceable descriptor record. Supporting evidence: Watanabe et al. [2] for validation-facing geometry control and del Jesus, Lara, and Losada [1] for separating impermeable and porous/dissipating assumptions.

8.11 Two- and Three-Dimensional Representation

8.12 Candidate Approaches

8.13 Critical Comparison

8.14 Assumptions and Applicability

8.15 Limitations and Failure Conditions

8.16 Project-Specific Conclusions

The Studentship shall not spend simulation effort on barrier shapes whose geometry cannot be reproduced, clipped to the corridor and post-processed for hydrodynamic metrics.

8.17 Selected Approach and Rationale

8.18 Unresolved Decisions

Open decisions are final mesh-quality limits, minimum wall clearance from artificial boundaries, and whether any dissipating concept is admissible without a porous closure.

8.19 Requirements and Handoffs

Software shall report a geometry-feasibility status before launching local runs, including corridor intersection checks, minimum cell-size estimates, wall clearance and unresolved-physics flags.

8.20 Deliverable Completion Record

9 RES-GEO-SPEC — Integrated Barrier-Class Geometry Specification

9.1 Purpose and Scope

This Deliverable consolidates the active fixed-rigid barrier geometry specification. It preserves the WBS ID while changing the active output away from geometry generation and toward reproducible wall-type and obstacle/dissipating/array comparisons in the two Tohoku corridors.

9.2 Research Questions

1. Are the active geometry classes, descriptors, placement rules and representations sufficient for local 3D hydrodynamic comparison?
2. Which fields must the Software workstream implement in the geometry manifest before data handling can proceed?
3. Which geometry ideas are explicitly deferred?

9.3 Terminology and Definitions

9.4 Literature Evidence

del Jesus, Lara, and Losada [1] supports the impermeable/opening-aware barrier distinction and limits unresolved porous claims. Watanabe et al. [2] supports preserving geometry and output identifiers for local impact validation. Mori, Takahashi, and The 2011 Tohoku Earthquake Tsunami Joint Survey Group [3] supports retaining the two Tohoku corridor settings as contrasting hydrodynamic contexts. Project conclusion: geometry specification is a data and modelling contract for comparison, not a literature survey or automatic design generator.

9.5 Governing Relations and Quantitative Definitions

The minimum geometry-case contract is:

$$\mathcal{C}_j^{\mathrm{geo}} = \left\{\texttt{corridor\_id},\, \texttt{geometry\_id},\, \mathcal{G}_j,\, \boldsymbol{\theta}_j^{\mathrm{geo}},\, \Gamma_{\mathrm{B},j},\, \mathcal{P}_j\right\} \tag{9.1}$$

where $\mathcal{G}_j$ is the descriptor record, $\boldsymbol{\theta}_j^{\mathrm{geo}}$ is the parameter vector, $\Gamma_{\mathrm{B},j}$ is the fixed barrier patch and $\mathcal{P}_j$ records provenance, CRS/datum linkage and unresolved physics flags. Source role: this equation integrates the decisions in RES-GEO-TAX, RES-GEO-PAR and RES-GEO-REP. Limitation: it does not define solver implementation or validation results.

9.6 Geometry Family or Representation

The active representation is fixed terrain plus fixed rigid barrier patches. Wall-type and obstacle/array-type cases share the same manifest structure so they can be compared through common hydrodynamic metrics.

9.7 Design Variables and Parameter Bounds

Only descriptor-level comparison variables are active. Generative control points, topology optimisation variables and learned geometry latents are deferred.

9.8 Geometric Descriptors

Minimum descriptors are class, position, orientation, crest elevation, height, width/thickness, footprint length, spacing, row count, opening fraction, roughness flag, blockage ratio and provenance quality.

9.9 Placement and Arrangement Rules

Placement shall be corridor-relative and shall preserve fixed measurement sections across class comparisons.

9.10 Feasibility Constraints

Feasibility requires terrain coverage, meshability, stable wall patch labels and no unresolved physics dependency for the active baseline.

9.11 Two- and Three-Dimensional Representation

The local three-dimensional representation is a fixed barrier surface used for no-slip/no-penetration boundary conditions and for pressure, shear, force, impulse and moment extraction.

9.12 Candidate Approaches

9.13 Critical Comparison

9.14 Assumptions and Applicability

9.15 Limitations and Failure Conditions

9.16 Project-Specific Conclusions

The integrated RES-GEO output now supports the immediate project scope: Tohoku/local comparison and barrier-class hydrodynamic comparison. It does not authorise geometry generation, structural damage simulation or ML surrogate training.

9.17 Selected Approach and Rationale

The active approach is descriptor-first fixed-rigid barrier-class comparison. Existing-defence reconstruction may be added where source geometry is available. Unrestricted generative geometry, topology optimisation, surrogate-led geometry search and deformable/damage geometry are deferred until hydrodynamic extraction and validation are stable.

9.18 Unresolved Decisions

- TODO: citable Kamaishi bay-mouth breakwater geometry and 2011 condition source.
- Final corridor-relative placement sections and geometry bounds.
- Geometry file format and HDF5/structured schema details.
- Roughness/opening/porous-closure activation policy.

9.19 Requirements and Handoffs

Software may begin the geometry-manifest MVP in parallel with Research: parse geometry case records, generate corridor-relative barrier patches, preserve patch identifiers, run clipping/meshability checks and export structured geometry metadata for replay-boundary and local 3D scaffolds. This is a data-handling handoff only, not solver implementation.

9.20 Deliverable Completion Record

References

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